

Functional Annotation Report: *fadE* (PP_1893, UniProt Q88LN6) in *Pseudomonas putida* KT2440

1. Summary / Answer to the Research Question

FadE (PP_1893, Q88LN6) is a membrane-anchored, FAD-dependent acyl-coenzyme A dehydrogenase (EC 1.3.8.7 / 1.3.8.8) that catalyzes the first and committed step of each cycle of fatty-acid β -oxidation: the α,β -dehydrogenation of a saturated medium- or long-chain fatty acyl-CoA to the corresponding *trans*-2(E)-enoyl-CoA, passing the abstracted electrons via its tightly bound FAD to electron-transfer flavoprotein (ETF) and onward to the respiratory chain. It works in the cytoplasm while tethered to the cytoplasmic (inner) membrane through an N-terminal transmembrane helix, and it operates within the fatty-acid degradation (β -oxidation) pathway together with FadD (acyl-CoA synthetase), FadB (multifunctional hydratase/dehydrogenase) and FadA (thiolase). In *P. putida* this pathway both feeds acetyl-CoA into central metabolism and supplies intermediates for medium-chain-length polyhydroxyalkanoate (mcl-PHA) biosynthesis.

Gene identity — verified. The symbol *fadE* is used correctly here. UniProt, KEGG ([ppu:PP_1893](#)), ortholog **K06445**, and eggNOG (**COG1960**) all annotate PP_1893 as an acyl-CoA dehydrogenase in the fatty-acid degradation pathway, and the protein belongs to the large (~815 aa, ~89 kDa) membrane-associated **FadE clade** typified by *Escherichia coli* FadE. This is distinct from the classic ~400-aa soluble mitochondrial ACADs and from *P. putida*'s own separate aromatic phenylacyl-CoA dehydrogenase (PP_0368). No conflicting-gene ambiguity was found; the naming, family, domains, and organism are all consistent.

2. Protein Identity and Domain Architecture

Property	Value	Source
UniProt	Q88LN6 (Q88LN6_PSEPK), protein existence: inferred from homology	UniProt
Gene / locus	<i>fadE</i> / PP_1893; genome position complement(2,134,303–2,136,750)	UniProt/KEGG
Length / mass	815 aa / 89,195 Da	UniProt
EC numbers	1.3.8.7 (medium-chain acyl-CoA dehydrogenase); 1.3.8.8 (long-chain)	UniProt
Cofactor	FAD (non-covalent)	UniProt (CHEBI:57692)
Orthology	KEGG K06445; eggNOG COG1960	KEGG/eggNOG
Domains	ACAD N-terminal (142–236), middle (240–338), C-terminal (363–508), bacterial-type C-terminal extension (517–802)	UniProt/Pfam (PF02771/PF02770/PF00441/PF09317)
Transmembrane	Helical TM segment, residues 43–64	UniProt

FadE is roughly twice the size of a canonical soluble ACAD. The extra bulk comes from **(i)** an N-terminal hydrophobic membrane anchor and **(ii)** a bacterial-type C-terminal extension (residues 517–802). This architecture is the hallmark of the *E. coli* FadE-type acyl-CoA dehydrogenases (COG1960), to which PP_1893 unambiguously belongs. **Orthology is firmly established:** *E. coli* FadE (gene *b0221*) is 814 aa and is assigned to the same KEGG ortholog group **K06445** as PP_1893 (815 aa) — an almost exact length match for this unusually large ACAD clade (KEGG). (A naïve global alignment to one shorter UniProt sequence record gave only ~28% identity; this reflects both the deep evolutionary divergence within this bacterial clade and a truncated sequence record, not a weak assignment — the KO/COG orthology call and the matching domain architecture and near-identical length are the decisive evidence.)

3. Primary Function: Reaction Catalyzed and Substrate Specificity

Reaction

FadE catalyzes the FAD-dependent α,β -dehydrogenation of a saturated fatty acyl-CoA thioester:

a (medium/long-chain) 2,3-saturated fatty acyl-CoA + oxidized [ETF] + H^+ $\rightarrow$ a (2E)-enoyl-CoA + reduced [ETF] (UniProt catalytic-activity annotations; Rhea RHEA:14477 for medium-chain, RHEA:17721 for long-chain)

This is **the first, committed step of every turn of the β -oxidation cycle**. In the *E. coli* and *Shewanella* orthologs this role is stated explicitly: FadE is "the first committed enzyme of the β -oxidation pathway" (Yusuf et al., 2020,

P 32507597

and "a flavoprotein involved in each turn of the beta-oxidation cycle" (Sirithanakorn & Imlay, 2024,

P 39436877

Catalytic mechanism (authoritative review)

Acyl-CoA dehydrogenases "constitute a family of flavoproteins that catalyze the α,β -dehydrogenation of fatty acid acyl-CoA conjugates" (Ghisla & Thorpe, 2004,

P 14728676

Mechanistically, an active-site glutamate base removes the substrate $\alpha C-H$ as a proton while the $\beta C-H$ is transferred "as a hydride to the flavin N(5) position," producing a reduced-enzyme-enoyl-CoA charge-transfer complex that is the point of electron transfer to ETF

P 14728676

This explains FadE's obligate dependence on FAD and its coupling to the ETF/respiratory electron sink rather than to NAD(P)⁺.

Substrate specificity

- UniProt assigns **both EC 1.3.8.7 (medium-chain) and EC 1.3.8.8 (long-chain)**, and KEGG's GenBank name is "medium-long chain acyl-CoA dehydrogenase" — i.e., a broad medium-to-long chain-length window rather than a single narrow substrate.
- Physiological requirement in the ortholog group: in *E. coli*, "Aerobic growth on sodium oleate was observed only for the cells with intact *fadE* gene" (Gulevich et al., 2016,

P 29369593

and catabolism of the medium-chain fatty acid dodecanoate (C12) proceeds through FadE

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These experiments establish that FadE-type activity is **necessary to initiate degradation of exogenous fatty acids as carbon/energy sources**.

- In *P. putida* KT2440, β -oxidation of exogenous medium/long-chain fatty acids (C6–C12) is the central catabolic route generating acetyl-CoA and supplying mcl-PHA monomers (Liu et al., 2023,

P 36763117

4. Subcellular Localization

FadE performs its chemistry **in the cytoplasm, while anchored to the cytoplasmic (inner) membrane**: - UniProt annotates a single helical **transmembrane segment at residues 43–64**; keywords include Membrane, Transmembrane, Transmembrane helix. - Kyte–Doolittle hydropathy analysis (this work) shows the protein's global hydrophobicity maximum (peak KD ≈ 2.6) over the N-terminal region (~residues 32–68), coincident with the annotated TM helix. - The catalytic ACAD domains (residues ~142–802) and FAD-binding site are soluble and cytoplasm-facing; GO assigns **C:cytoplasm** (GO:0005737), **F:FAD binding** (GO:0050660), **F:long-chain** (GO:0004466) and **medium-chain** (GO:0070991) **acyl-CoA dehydrogenase activity**, and **P:fatty acid β -oxidation using acyl-CoA dehydrogenase** (GO:0033539).

Thus FadE is best described as a **monotopic/peripheral inner-membrane flavoenzyme** whose N-terminal helix physically couples the first oxidation of β -oxidation to the membrane-proximal ETF/electron-transport machinery. This distinguishes the bacterial FadE clade from the fully soluble matrix ACADs of mitochondria.

5. Pathway Context, Partners, and Regulation

The β -oxidation spiral in *P. putida* KT2440

FadE is step 1 of the four-reaction cycle that shortens an acyl chain by two carbons per turn: 1. **Activation** — free fatty acid $\rightarrow$ acyl-CoA, by FadD (long-chain-fatty-acid-CoA ligase; *fadD-I/II* = PP_4549/PP_4550, EC 6.2.1.3). 2. **Dehydrogenation** — acyl-CoA $\rightarrow$ *trans*-2-enoyl-CoA, by **FadE (PP_1893)** $\leftarrow$ *this gene*. 3. **Hydration + oxidation** — enoyl-CoA $\rightarrow$ 3-hydroxyacyl-CoA $\rightarrow$ 3-ketoacyl-CoA, by the multifunctional FadB (PP_2136; EC 4.2.1.17/1.1.1.35). 4. **Thiolysis** — 3-ketoacyl-CoA $\rightarrow$ acetyl-CoA + (chain-2)acyl-CoA, by FadA thiolase (PP_2051; EC 2.3.1.16).

The released acetyl-CoA enters the TCA cycle; in *P. putida* the (R)-3-hydroxyacyl-CoA intermediates are also channelled (via PhaJ enoyl-CoA hydratases) into mcl-PHA polyester storage

(P 36763117).

Genomic organization and paralog redundancy

PP_1893 is a **standalone gene**, not part of the *fadBA* operon — it is flanked by an unrelated type-1 fimbrial cluster (*fimD/fimC/fimI* = PP_1889–1891), a hypothetical ORF (PP_1892), and a glutathione-S-transferase/ABC-transporter region (PP_1894–1896) (KEGG genome map). PP_1893 is the genome's **only** K06445/COG1960 FadE-type ACAD, but *P. putida* additionally encodes several other acyl-CoA dehydrogenases — a short-chain enzyme *acd* (PP_2216, EC 1.3.8.1), three EC 1.3.8.7 enzymes (PP_2437, PP_2793, PP_3725; K00249), and the inducible aromatic phenylacyl-CoA dehydrogenase PP_0368 (McMahon & Mayhew, 2007,

P 17559393).

This paralog set implies **partial functional redundancy** in the dehydrogenation step, consistent with the robustness of β -oxidation in this organism.

Regulation (characterized in the *E. coli* ortholog)

In *E. coli*, *fadE* (originally *yafH*) is a genuine acyl-CoA dehydrogenase whose transcription is controlled by the **FadR** repressor (Sadovskaya et al., 2001,

P 11771124).

The *fadE* operon is induced ~10-fold by long-chain fatty acids (C12 partial, $\geq$ C14 full), repressed by glucose, and constitutively expressed in *fadR* mutants (Clark, 1981, [P 6271734](#));

long-/medium-chain acyl-CoA effectors bind FadR to relieve repression, co-inducing *fadE* and *fadBA* (Zhang et al., 2006, [P 16734787](#)).

An analogous fatty-acid-responsive control is expected in *P. putida* (which carries FadR/PsrA-type regulators), though this specific regulation has been demonstrated experimentally mainly in the enterobacterial model.

6. Evidence Basis

- **Direct experimental (orthologs):** genetic requirement of *fadE* for aerobic growth on oleate and role in dodecanoate catabolism in *E. coli* (PMIDs 29369593, 39436877); mechanistic enzymology of ACADs

([P 14728676](#));

FadR-mediated transcriptional control (PMIDs 6271734, 16734787, 11771124); substrate-differentiated FadE homologs in *Shewanella*

([P 32507597](#)).

- **Organism-specific (*P. putida* KT2440):** central role of β -oxidation and its coupling to mcl-PHA metabolism

([P 36763117](#));

existence of a separate large aromatic-acyl-CoA dehydrogenase, PP_0368

([P 17559393](#)).

- **Database / bioinformatic:** UniProt Q88LN6 (EC, catalytic-activity Rhea reactions, FAD cofactor, TM helix 43–64, domain map, GO); KEGG K06445 / module M00087 / pathway ppu00071 and genome neighborhood; eggNOG COG1960; Kyte–Doolittle hydropathy (this work) confirming the N-terminal membrane anchor.

7. Supported and Refuted Hypotheses

Supported - H1: FadE is an FAD-dependent acyl-CoA dehydrogenase catalyzing β -oxidation's first, committed dehydrogenation. ✓ (UniProt/KEGG + ortholog literature) - H2: Broad medium/long-chain substrate specificity. ✓ (dual EC assignment; KEGG name; ortholog growth phenotypes) - H3: Cytoplasm-facing enzyme anchored to the inner membrane by an N-terminal TM helix. ✓ (UniProt features + GO + hydropathy) - H4: Functions in the β -oxidation pathway with FadD/FadB/FadA; feeds acetyl-CoA and mcl-PHA precursor supply. ✓ (KEGG pathway;

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Refuted / not supported - That *fadE* lies in the *fadBA* operon — **refuted**: PP_1893 is a standalone gene among fimbrial/transporter genes. - That PP_1893 is the sole acyl-CoA dehydrogenase / non-redundant — **not supported**: multiple paralogous ACADs exist (PP_2216, PP_2437, PP_2793, PP_3725, PP_0368).

8. Limitations and Future Directions

- **No enzyme-level study of PP_1893 itself** exists; its precise chain-length optimum, kinetic parameters (kcat/Km across C4–C18 acyl-CoAs), and quaternary structure are inferred from orthology and family membership, not measured directly. Direct biochemistry on purified PP_1893 would define its true substrate profile.
- **Regulation in *P. putida*** (FadR/PsrA involvement, inducer chain-length threshold) has not been experimentally mapped for PP_1893 specifically.
- **Redundancy** among the *P. putida* ACADs means the phenotypic consequence of a *fadE* single knockout is untested; combinatorial deletions would clarify which paralogs cover which chain lengths.
- Localization is inferred from sequence/GO; membrane fractionation or fluorescent tagging would confirm inner-membrane association experimentally.

Prepared through iterative analysis of UniProt, KEGG, eggNOG/InterPro annotations, sequence/hydropathy analysis, and primary literature on FadE-clade acyl-CoA dehydrogenases and P. putida β -oxidation.

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